MASTER OF BUSINESS ADMINISTRATION

SEMESTER 3

DADS304

VISUALIZATION

Unit 2

Basic Visualization Using R

Table of Contents

SL No	Topic	Fig No / Table / Graph	SAQ / Activity	Page No
1	Introduction	-	-	3
	1.1 Objectives	-	-	
2	Features of R Software	-	-	4
3	Steps to Install R	1 to 9	-	5 - 9
4	Basic commands using R	10, 11, 12	1	10 -11
5	Basic Statistical Commands using R	13 to 14	2	12
6	Basic Plotting using R	15 to 22	3	13 -17
7	Advanced Plotting using R	23 to 30	4	18 - 21
8	Activities	-	-	22
9	Summary	-	-	23
10	Glossary	-	-	23
11	Study Notes and Did You Know	-	-	24
12	Case Study	-	-	24
13	Terminal Questions	-	-	25
14	Answers Self Assessment Questions	-	-	25
15	Terminal Answer Key	-	-	26 - 28
16	Concept map	-	-	29
17	References	-	-	29

1. INTRODUCTION

There is an enormous amount of data available in the market. In recent times, online platform utilization is more than offline platforms. Data are getting stored in multiple formats and redundancy of the data is also very high. It is getting incremented exponentially day by day. If it keeps progressing like this, one day there will be a huge storage crisis. This problem should be addressed immediately. Data visualization is the process through which raw data can be visualized and some inference can be derived and based on that inference, other methods of data pre-processing can be applied. The pre-processed data is the ready data made available for further analysis to get proper inference from the data. It is in turn helpful in clearing the storage to a certain extent, so that it can be utilized further in an effective manner. There are many platforms to visualize data. In this unit, R software will be used for data visualization, as R software is a free open-source software with GNU package. R is a statistical programming language where data can be visualized from a basic to advanced level before and after statistical modelling.

1.1 Objectives:

After studying this unit, you should be able to:

- *Identify basic commands using R*
- *Explain basic statistical commands using R*
- *Discuss the basics of data visualization using R*
- *Identify the advanced data visualization using R*

2. FEATURES OF R SOFTWARE

A few major features and roles of R software are as follows:

2.1 Open Source

R uses open-source software environment, where packages can be added and removed as per the requirements of the applications and projects. It is available free of cost.

2.2 Strong Graphical Environment

R can create an environment of static graphics as well as dynamic interactive graphical atmosphere. It helps in data visualization as well as data representation.

2.3 Exclusive Active Community Support

R has strong open-source library. There are a huge number of contributors for the development of R libraries which provides strong community support.

2.4 A Wide Selection of Packages

Comprehensive R Archive Network has more than 10,000 packages, which can help the users solve multidimensional problem solving.

2.5 Comprehensive Environment

R is an object-oriented programming language. It has an exclusive robust package, ,Rshiny, which can be used to develop Web apps.

2.6 Cross Platform Support

R is machine independent. It can run on any operating system and can support cross platform activities.

2.7 Support Other Programming Languages

R can support other programming languages. Many of the functions in R are written with C, C++, Fortran.

2.8 Advanced Statistical Calculation

R supports basic statistical calculations to advanced statistical calculations to get proper and effective inference from the data.

3. STEPS TO INSTALL R

Step 1: Go to the website The Comprehensive R Archive Network (r-project.org)

Fig. 2.1: CRAN Project Website

Step 2: Download R as per operating system availability in the system. Follow the below links as per OS availability.

https://cran.r-project.org/bin/linux/

https://cran.r-project.org/bin/macosx/

https://cran.r-project.org/bin/windows/

Step 3: Install R on Windows.

Step 4: Click on Base Subdirectory Link.

Step 5: Click download R version for windows.

Fig. 2.2: Download R Version

Step 6: Run .exe file and follow the instruction.

Step 6a: Select the desired language and click next.

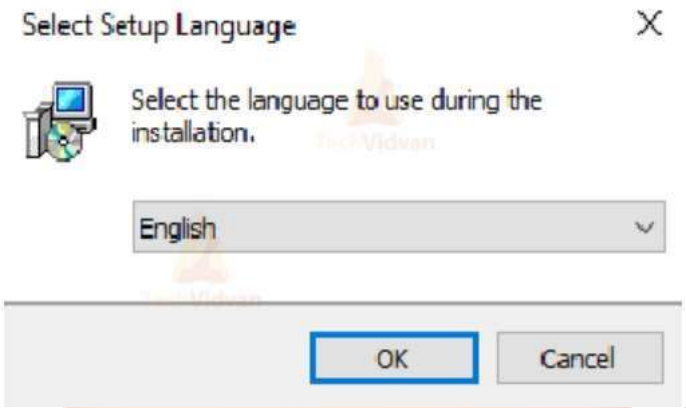


Fig. 2.3: Select the desired language

Step 6b: Read the license agreement and click next.

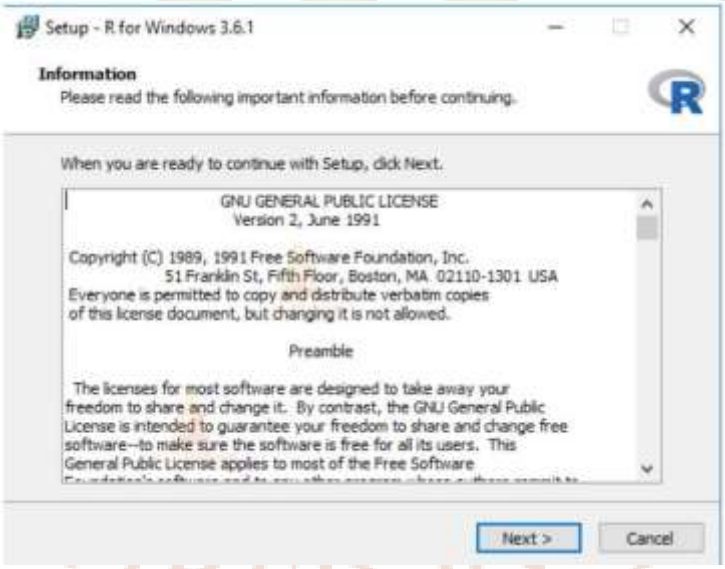


Fig. 2.4: License Agreement

Step 6c: Select the components you wish to install and click next.

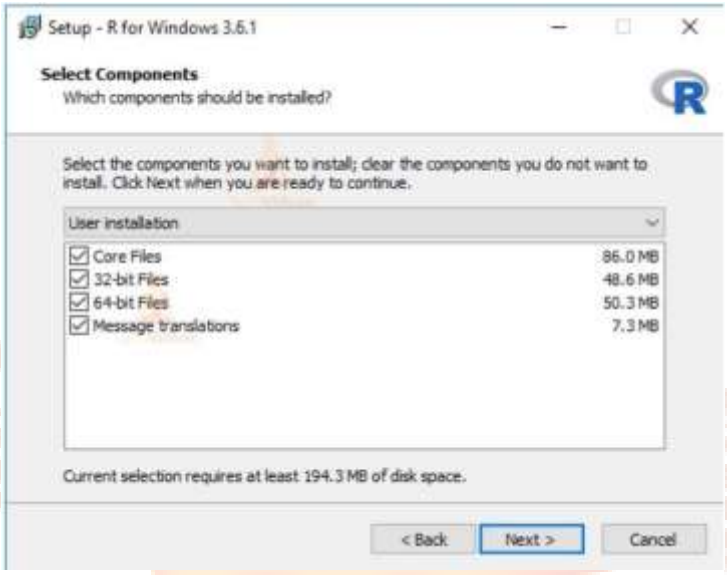


Fig. 2.5: Select Components

Step 6d: Enter the path you want to install R at.

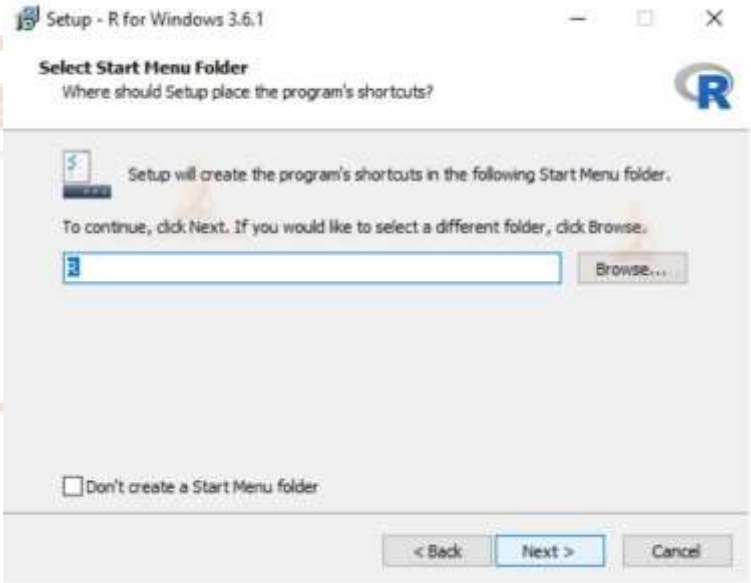


Fig. 2.6: Selection of Path

Step 6e: Select Additional Tasks.

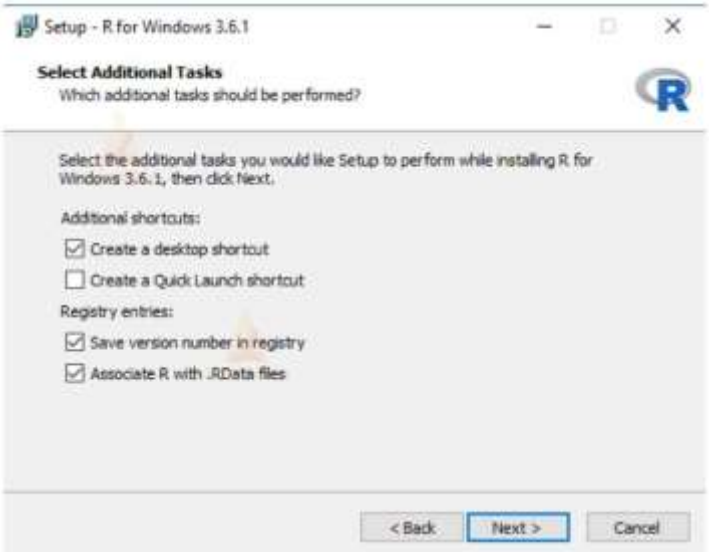


Fig. 2.7: Selection of Additional Task

Step 6f: Installation Process begins.

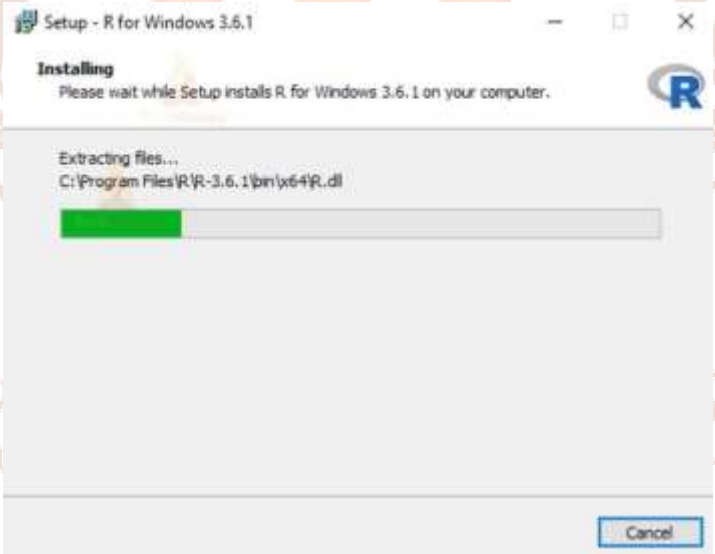


Fig. 2.8: Installation Process

Step 6g: Click on Finish Installation to complete the process.

Fig. 2.9: Complete Installation Process

4. BASIC COMMANDS USING R

This part deals with basic R commands. The R console can be visualized as

Fig. 2.10: R Console

1. Basic Command to print Welcome

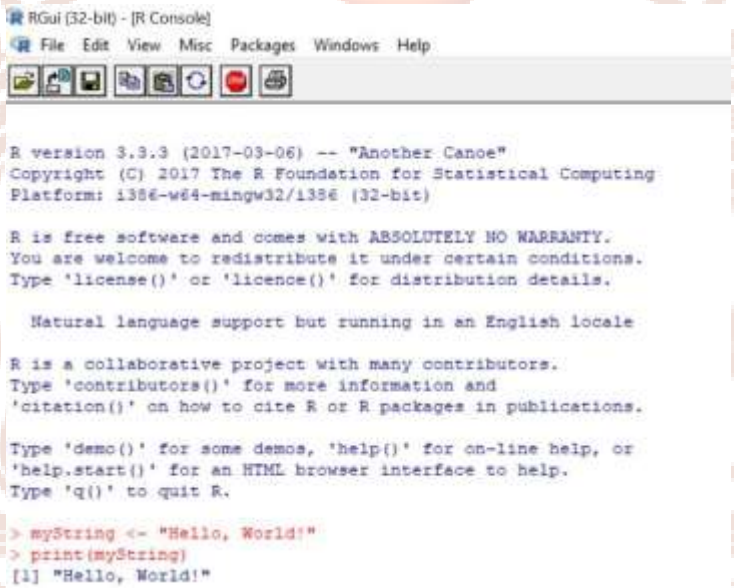


Fig. 2.11: Print Command

2. c () – Enter data manually to a vector in R.

```
> x = c(8, 6, 7, 5, 3, 0, 9)
> x

[1] 8 6 7 5 3 0 9

> names = c("Owen", "Luke", "Anakin", "Leia", "Jacen", "Jaina")
> names

[1] "Owen"   "Luke"   "Anakin" "Leia"   "Jacen"  "Jaina"
```

Fig. 2.12: c()Command

3. data () – Load (often into a data. frame) built-in dataset
4. dim () – Get or Set dimension of the specified built-in dataset
5. names () – Lists names of variables in a data. frame
6. View () – Lists names of variables in a data. frame
7. Str () – Display internal structure of an R object

```
> data(iris)
> names(iris)
[1] "Sepal.Length" "Sepal.Width"  "Petal.Length" "Petal.Width"  "Species"
> dim(iris)
[1] 150   5
> str(iris)
'data.frame':	150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species     : Factor w/ 3 levels "setosa","versicolor",..: 1 1 1 1 1 1 1 1 1 1 ...
> View(iris)
```

Fig. 2.13: data (), name (), dim (), str (), View () Command

8. read.csv () – Used to read the .csv files
9. read. Table () – Used to read tables

Self-Assessment Questions - 1

1. _____command lists names of variables in a data. frame.
2. _____ command used to read .csv files.

5. BASIC STATISTICAL COMMANDS USING R

R is famous for statistical operations. There are multiple in-built statistical functions in R.

1. mean () – Used to identify mean of the data
2. median () – Used to identify median of the data
3. summary () – Used to get the summary of the data
4. var () – Used to identify variance of the data
5. sd () – Used to identify standard deviation of the data
6. quantile () – Used to identify quantile of the data

```
> counties=read.csv("http://www.calvin.edu/~stob/data/counties.csv")
> names(counties)
[1] "County"        "State"         "Population"    "HousingUnits"  "TotalArea"
[6] "WaterArea"     "LandArea"      "DensityPop"    "DensityHousing"
> x = counties$LandArea
> mean(x, na.rm = T)
[1] 1126.214
> median(x, na.rm = T)
[1] 616.48
> summary(x)
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.99  431.70  616.50 1126.00  923.20 145900.00
> sd(x, na.rm = T)
[1] 3622.453
> var(x, na.rm = T)
[1] 13122165
> quantile(x, probs=seq(0, 1, .2), na.rm=T)
      0%      20%      40%      60%      80%     100%
    1.99   403.29   554.36   717.94  1043.82 145899.69
```

Fig. 2.14: mean (), median (), summary (), var (), sd (), quantile () Command

Self-Assessment Questions - 2

3. _____ command is used to summarize the data.
4. sd () command is used for _____ .

6. BASIC PLOTTING USING R

1. Bar Plot: Bar Plot conveys relational information.

```
barplot()

pol = read.csv("http://www.calvin.edu/~stob/data/csbv.csv")
barplot(table(pol$Political04), main="Political Leanings, Calvin Freshman 2004")
barplot(table(pol$Political04), horiz=T)
barplot(table(pol$Political04),col=c("red","green","blue","orange"))
barplot(table(pol$Political04),col=c("red","green","blue","orange"),
        names=c("Conservative","Far Right","Liberal","Centrist"))
```

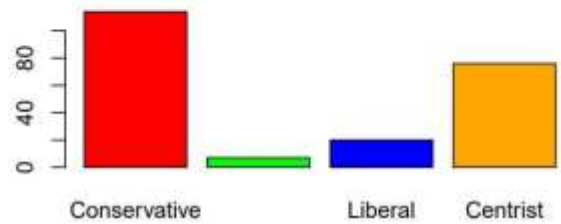


Fig. 2.15: Bar Plot

2. Box Plot: It is used to demonstrate locality, spread and skewness of the data.

```
boxplot()

data(iris)
boxplot(iris$Sepal.Length)
boxplot(iris$Sepal.Length, col="yellow")
boxplot(Sepal.Length ~ Species, data=iris)
boxplot(Sepal.Length ~ Species, data=iris, col="yellow", ylab="Sepal length",main="Iris Sepal Length by Species")
```

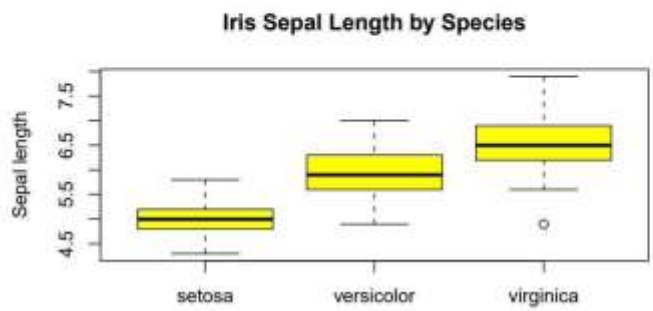


Fig. 2.16: Box Plot

3. Scatter Plot: It explains relationship between two variables in a data set.

```
plot()

data(faithful)
plot(waiting~eruptions,data=faithful)
plot(waiting~eruptions,data=faithful,cex=.5)
plot(waiting~eruptions,data=faithful,pch=6)
plot(waiting~eruptions,data=faithful,pch=19)
plot(waiting~eruptions,data=faithful,cex=.5,pch=19,col="blue")
plot(waiting~eruptions, data=faithful, cex=.5, pch=19, col="blue", main="Old Faithful Eruptions",
     ylab="Wait time between eruptions", xlab="Duration of eruption")
```

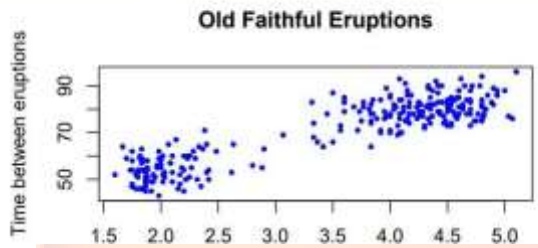


Fig. 2.17: Scatter Plot

4. Histogram: It is used to summarize continuous or discrete data in an interval scale.

Hist ()

```
> par(mfrow = c(1,2))      # set figure so next two plots appear side-by-side
> poisSamp = rpois(50, 3)    # Draw sample of size 50 from Pois(3)
> maxX = max(poisSamp)     # will help in setting horizontal plotting region
> hist(poisSamp, freq=F, breaks=-.5:(maxX+.5), col="green", xlab="Sampled values")
> plot(0:maxX, dpois(0:maxX, 3), type="h", ylim=c(0,.25), col="blue", main="Probabilities for Pois(3)")
```

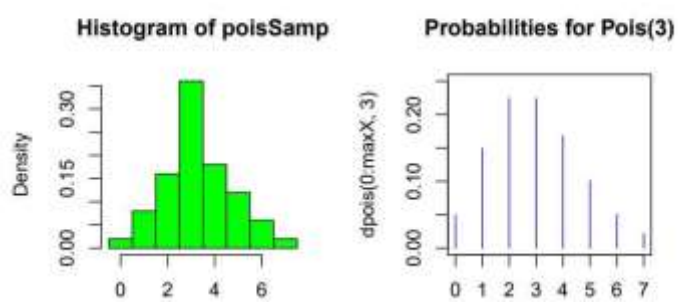


Fig. 2.18: Histogram

5. pnorm (), qnorm (): pnorm is cumulative density function of the normal distribution. qnorm finds the boundary value.

   pnorm (), qnorm ()

```
> pnorm(17, 19, 3)     # gives Prob[X < 17], when X ~ Norm(19, 3)
[1] 0.2524925
> qnorm(c(.95, .975, .995))       # obtain z* critical values for 90, 95, 99% CIs
[1] 1.644854 1.959964 2.575829
> nSamp = rnorm(10000, 7, 1.5)    # draw random sample from Norm(7, 1.5)
> hist(nSamp, freq=F, col="green", main="Sampled values and population density curve")
> xs = seq(2, 12, .05)
> lines(xs, dnorm(xs, 7, 1.5), lwd=2, col="blue")
```

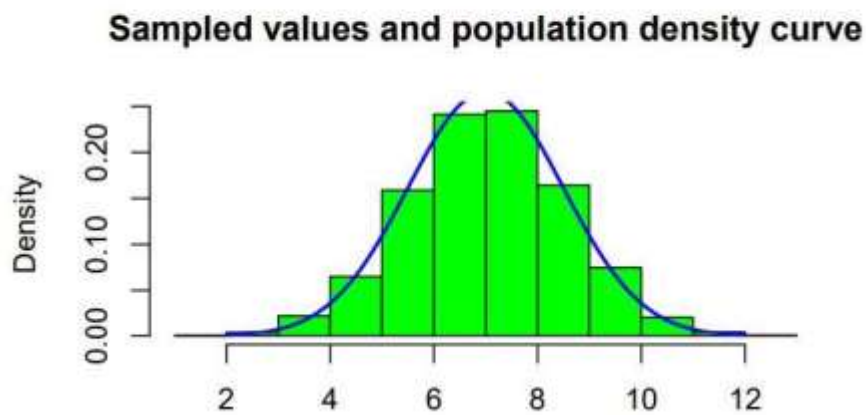


Fig. 2.19: pnorm (), qnorm ()

6. Line Graph: It is used to plot lines as per the relationships between two variables.

Plot()

```
# Create the data for the chart.
v <- c(7,12,28,3,41)

# Give the chart file a name.
png(file = "line_chart.jpg")

# Plot the bar chart.
plot(v,type = "o")

# Save the file.
dev.off()
```

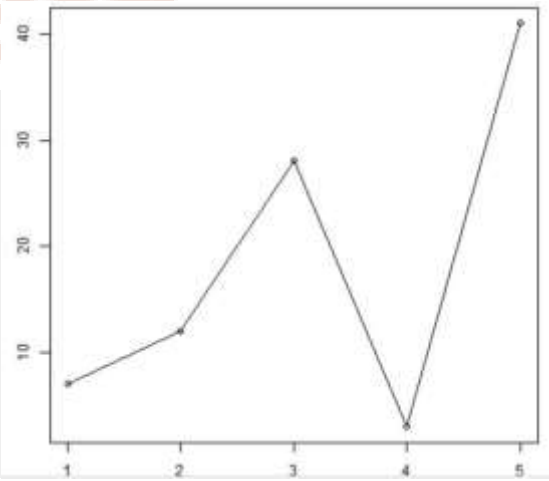


Fig. 2.20: Line Graph

7. Pie Chart: Pie chart is used to plot percentage distribution of the data.

Pie ()

```
# Create data for the graph.
x <- c(21, 62, 10, 53)
labels <- c("London", "New York", "Singapore", "Mumbai")

# Give the chart file a name.
png(file = "city.png")

# Plot the chart.
pie(x,labels)

# Save the file.
dev.off()
```

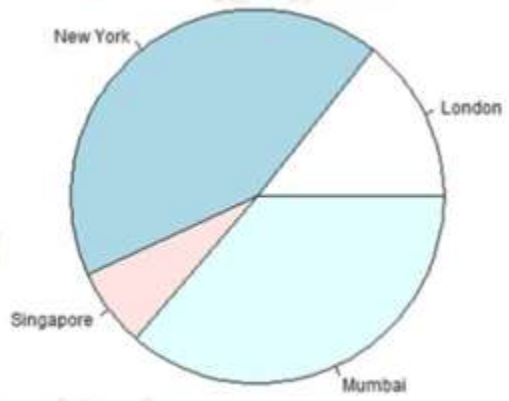


Fig. 2.21: Pie Chart

8. Stacked Bar Graphs: Data visualization of bar charts use horizontal columns to exhibit numerical comparisons between categories.

```
dat <- read.table(text = "A   B   C   D   E   F    G
1 10 80 30 90 70 60  90
2 20 50 70 50 40 10  40
3 60 80 80 60  60 30 160
4 20 40 70 80 20 10  70", header = TRUE)

barplot(as.matrix(dat))
```

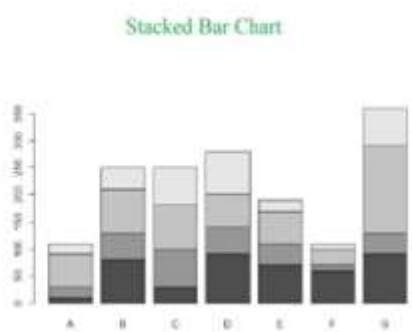


Fig. 2.22: Stacked Bar Graphs

Self-Assessment Questions -3

5. _____ is a chart used to visualize a large data by connecting points in a continuous line.
6. _____ describes the distribution of data over a continuous interval or particular period of time.

INSPIRED BY LIFE

7. ADVANCED PLOTTING USING R

Data visualization is very important to identify the trend of the data. There are different types of plots to get inference from the data. Libraries are needed to be included for advanced plotting using R.

1. ggplot2: It is a plotting package which helps to create complex plots from various data in a data frame. It is a more advanced plot to get programmatic interface, clear visibility and proper inference.

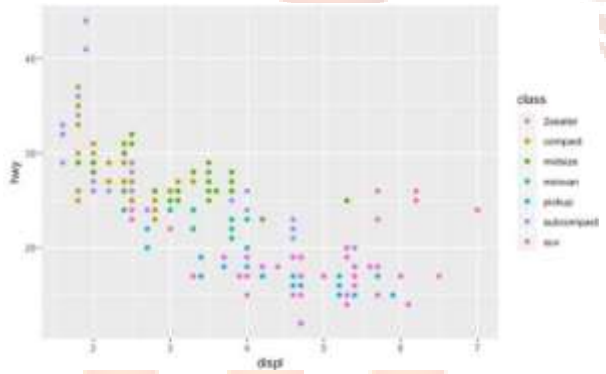


Fig. 2.23: ggplot2

2. Lattice: It is a package which has graphics and data visualization property that originated from Trellis graphics package. It can plot multivariate data. It can first plot basic visualization and advance it based on enhanced features.

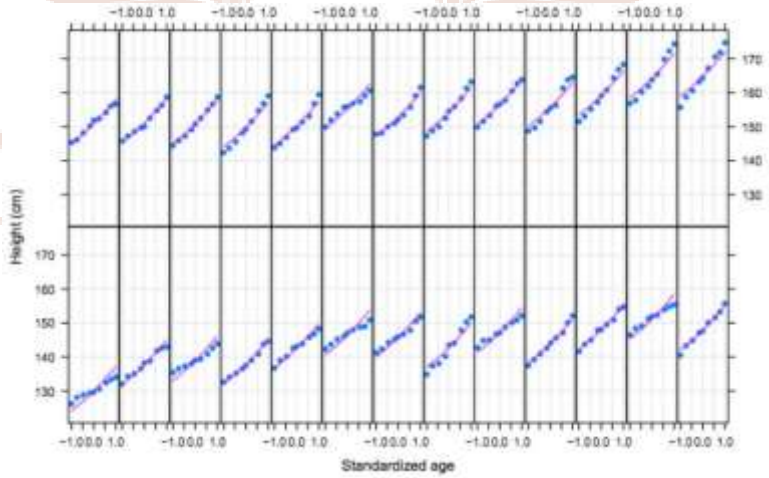


Fig. 2.24: Lattice

3. High charter: It is associated with java script library and its modules. It is highly flexible and customizable. It has a very high powerful API. Chart visualization of the data is possible using this.

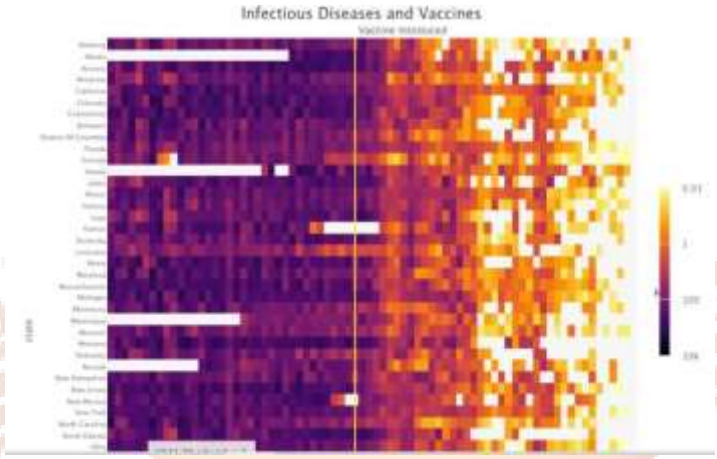


Fig. 2.25: High Charter

4. Leaflet: It is an open-source java script library used to create dynamic online maps. It can create graph through layer wise.

Fig. 2.26: Leaflet

5. R Color Brewer: It is an important tool for color management. It offers several color pallets and provides some unique graphical visualization.

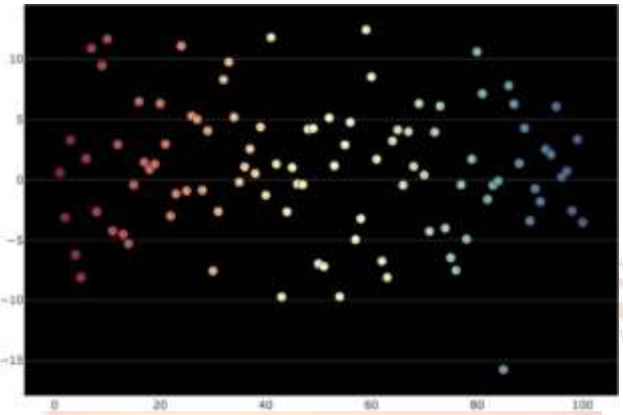


Fig. 2.27: R Color Brewer

6. Plotly: It is a R package which can help create interactive web services with the help of java scripts. It is an open-source library.

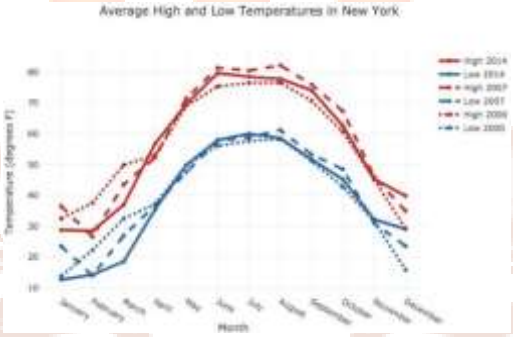


Fig. 2.28: Plotly

7. Sunburst R: It is a special type of data visualization tool in R. It is customizable.

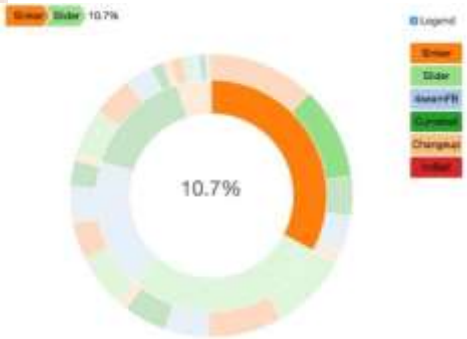


Fig. 2.29: Sunburst R

8. RGL: It is used to produce interactive 3-D plots. It contains high level graphic commands along with basic commands. It is used for 3-D visualization with openGL.

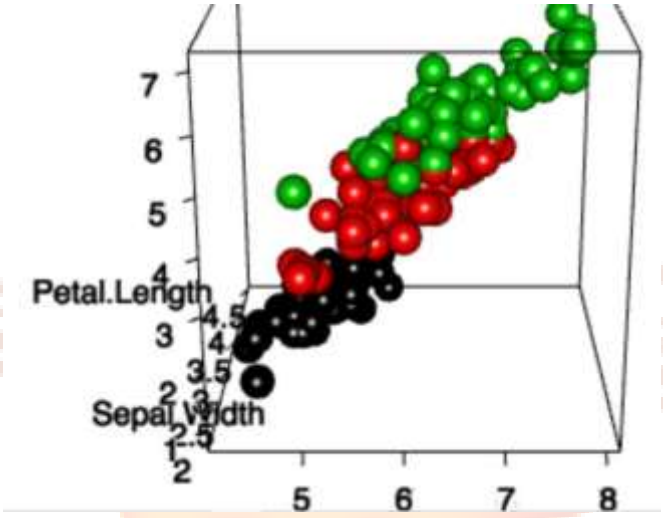


Fig. 2.30: RGL

9. Dygraphs: It is a java script charting library. It creates high facilities of time series data using R.

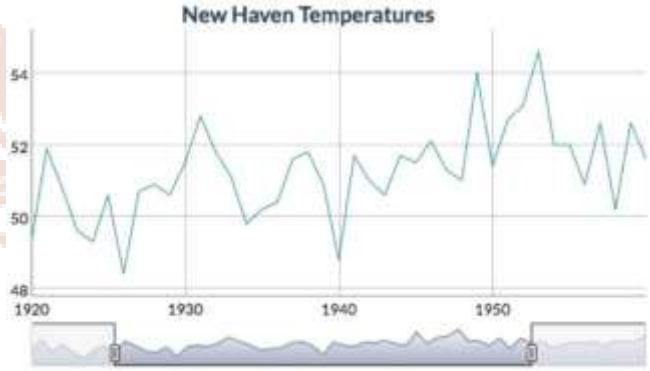


Fig. 2.31: Dygraphs

Self-Assessment Questions -4

7. _____ which has graphics and data visualization property originated from Trellis graphics package.
8. _____ produce interactive 3-D plot.

8. ACTIVITIES

Activity A

Create a database of cancer patients with proper feature set and infer using data visualization techniques.

……

…… Activity:

Identify the major symptoms of cancer and its stages and create a data set with minimum 100 cases. Visualize the data and get inference.

Activity B

Visualize Titanic dataset using data visualization command in R.

……

…… Activity:

Use the link Titanic Dataset | Kaggle

9. SUMMARY

Data visualization is a major technology for data analysis. Data is a very important asset of a company. But at the same time repetitive and useless data creates unnecessary storage overflow and leads to wrong decision-making. There are multiple tools and techniques available for data visualization such as tableau, Microsoft Power BI, Python, Microsoft Excel, Mongo DB, R Studio and many more. The visualization techniques are more or less the same for all the cases. The main challenge for data visualization is to identify proper visualization technique for respective datasets. For example, if it is required to display data based on percentage, pie chart is the best option. Box plot is used to identify spread and skewness of the data. There are advanced plotting techniques where libraries are needed to be added to R and proper inferences from the data is possible. However, before undertaking any data visualization through R, the basic commands used in the platform of R are required to be understood. In this unit, the basic R commands, essential statistical commands, basic and advanced data visualization techniques were discussed.

10. GLOSSARY

Braided Graph: a novel visualization technique where filled areas are sorted in depth order.

Bullet Graphs: used for comparing the performance of primary measures with other measures.

Error Bars: used to identify estimated error in data.

Marimekko Charts: used to visualize categorical data over a pair of variables.

11. STUDY NOTES & DID YOU KNOW

Did You Know? About 65% of the brands use infographics for marketing. 84% of the people accepted infographics as a powerful tool. Infographics are the fourth most used type of content marketing. Infographics can boost website traffic by 12%.

Did You Know? 80% of the health care market players invested in big data analytics and artificial intelligence driven by market demand.

12. CASE STUDY

Collect the Retinopathy database from your nearby health care center and visualize the cases of positive, negative.

Q1. Identify the percentage of positive cases in your area.

……

Q2. Identify the percentage of negative cases in your area.

……

13. TERMINAL QUESTIONS

Short Answer Type Questions:

1. Briefly discuss about data visualization tools.
2. What is the use of box plot in data visualization?
3. What is the importance of R for data visualization in data science?

Long Answer Type Questions:

1. Explain the various basic commands in R.
2. Explain the R commands for basic data visualization.
3. Explain the R commands for advanced data visualization.

14. ANSWERS-SELF-ASSESSMENT QUESTIONS

1. View ()
2. read.csv ()
3. summary ()
4. Used to identify standard deviation of the data
5. A line graph
6. Histograms
7. Lattice
8. RGL

15. TERMINAL ANSWER KEY

Short Answer Type Questions Answer Key

1. Briefly discuss about data visualization tools.

   Ans: There are many data visualization tools available in the market. Some features in them are common. It is a software that can be used to visualize the data. Data visualization is very important for the purpose of data analysis. The most popular data visualization tools are tableau, Microsoft excel, Microsoft power BI, Dundas BI, Jupiter, Zoho reports, Google Charts, Visual.ly, RAW, IBM Watson, Sisense, Plotly, data wrapper, fusion charts, Qlik view, info grams, chart blocks, D3.js, Chart.js, chartist.js, sigma.js, and polymaps.

2. What is the use of box plot in data visualization?

   Ans: Box plots are used to visualize the spread of data. If the data values are beyond the maximum and minimum boundary values, those are known as outliers and cannot be considered to calculate the mean of the data. The median will be at the middle and the full data sets are divide into four quartiles. The difference between lower and upper quartile is known as Inter Quartile Range (IQR).

3. What is the importance of R for data visualization in data science?

   Ans: There are several basic and advanced tools are available for data visualization using R. So, it is very easy to visualize direct data and /or processed data after statistical processing. R is a strong statistical tool for data visualization.

Long Answer Type Questions Answer Key

1. Explain the various basic commands and statistical commands in R.

   Ans:

   Basic Commands are as follows:

   1. print () – to print

2. c ()– To enter data manually to a vector in R
3. data () – To load (often into a data. frame) built-in dataset
4. dim () – To load (often into a data. frame) built-in dataset
5. names () – Lists names of variables in a data. frame
6. View () – Lists names of variables in a data. frame
7. Str () – Displays internal structure of an R object
8. read.csv () – Used to read the .csv files
9. read. Table () – Used to read tables
10. mean () – Used to identify mean of the data
11. median () – Used to identify median of the data
12. summary () – Used to get the summary of the data
13. var () – Used to identify variance of the data
14. sd () – Used to identify standard deviation of the data
15. quantile () – Used to identify quantile of the data

2. Explain the basic data visualization using R.

Ans:

1. Bar Plot: Bar Plot conveys relational information.
2. Box Plot: It is used to demonstrate locality, spread and skewness of the data.
3. Scatter Plot: It explains relationship between two variables in a data set.
4. Histogram Hist() : It is used to summarize continuous or discrete data in an interval scale.
5. pnorm (), qnorm (): pnorm is cumulative density function of the normal distribution. qnorm finds the boundary value.
6. Line Graph Plot() : It is used to plot lines as per the relationships between two variables.
7. Pie Chart Pie(): Pie chart is used to plot percentage distribution of the data.
8. Stacked Bar Graphs: Data visualization of bar charts use horizontal columns to exhibit numerical comparisons between categories.

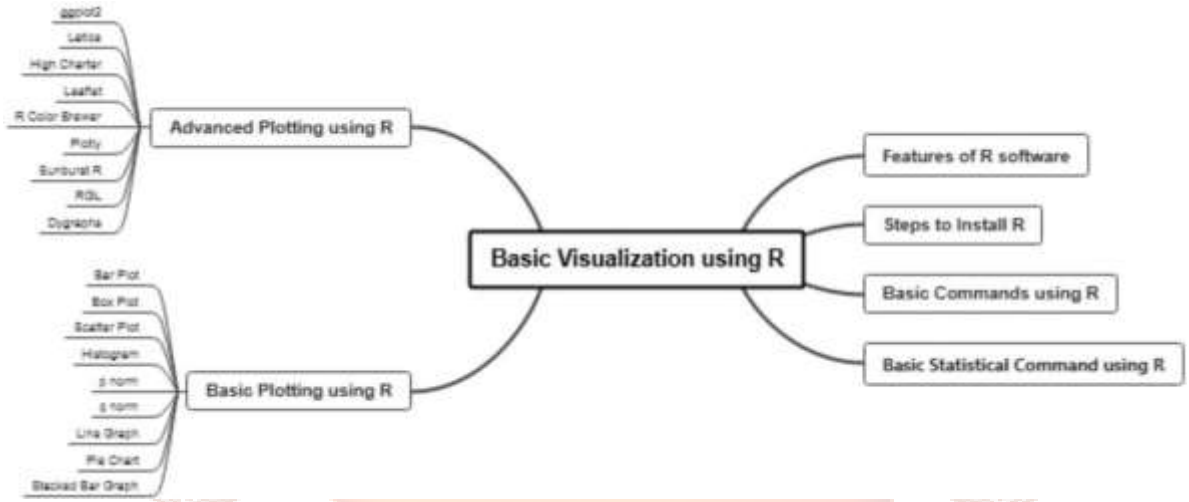
3. Explain advanced data visualization using R.

Ans:

Data visualization is an important technique to identify the trends in the data. There are different types of plots to get inference from the data. Libraries are needed to be included for advanced plotting using R.

1. ggplot2: It is a plotting package which helps to create complex plots from various data in a data frame. It is a more advanced plot to get programmatic interface, clear visibility and proper inference.
2. Lattice: It is a package which has graphics and data visualization property that originated from Trellis graphics package. It can plot multivariate data. It can first plot basic visualization and advance it based on enhanced features.
3. High charter: It is associated with java script library and its modules. It is highly flexible and customizable. It has a very high powerful API. Chart visualization of the data is possible using this.
4. Leaflet: It is an open-source java script library, used to create dynamic online maps. It can create graph through layer wise.
5. R Color Brewer: It offers several color pallets. It is an important tool for color management. It offers several color pallets and provides some uniqueness in graphical visualization.
6. Plotly: It is a R package which can help to create interactive web services with the help of java scripts. It is an open-source library.
7. Sunburst R: It is a special type of data visualization tool in R. It is customizable.
8. RGL: It is used to produce interactive 3-D plots. It contains high level graphic commands along with basic commands. It is used for 3-D visualization with open GL.
9. Dygraphs: It is a java script charting library. It creates high facilities of time series data using R.

16. CONCEPT MAP



17. REFERENCE

L. S. Hovhan, *7 Key Benefits of Interactive Data Visualization*, October 2020, [online] Available: https://infogram.com/blog/7-key-benefits-of-interactive-data-visualization/.

Likhitha Ravi, E. Kauffmann, J. Peral, D. Gil, A. Ferrández, R. Sellers and H. Mora, "A framework for Big Data Analytics in Commercial Social Networks: A Case Study on Sentiment Analysis and Fake Review Detection for Marketing Decision-Making", *Industrial Marketing Management*, vol. 90, pp. 523–537, 2020.

K. Govindan and H. Gholizadeh, "Robust Network Design for Sustainable-Resilient Reverse Logistics Network using Big Data: A Case Study of end-of-life vehicles", *Transportation Research Part E: Logistics and Transportation Review*, vol. 149, pp. 102279, 2021.

E-References

https://onlinecourses.nptel.ac.in/noc22_mg09/

NOC | Essentials of Data Science With R Software - 1: Probability and Statistical Inference (nptel.ac.in)

NPTEL

https://onlinecourses.nptel.ac.in/noc19_ma33